

# SEMI-SUPERVISED SEGMENTATION USING A CROSS-TEACHING ENSEMBLE MODEL

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## ABSTRACT

The scarcity of labeled data limits the effectiveness of fully supervised segmentation models on complex, newer datasets. This experiment evaluates a semi-supervised cross-teaching framework combining a U-Net and a Vision Transformer (ViT) trained on the Oxford-IIIT Pet dataset. Using only 20% of available labels, the ensemble cross-teaching model outperforms both individual models on every evaluation metric. The results demonstrate that cross-teaching is well-suited for largely unlabeled datasets.

## 1. INTRODUCTION

Deep learning, particularly encoder-decoder architectures like U-Net, has become a standard approach for image segmentation across scientific imaging and computer vision. It enables automated detection of fine structural features while preserving spatial detail, and serves as a backbone for applications ranging from medical imaging to autonomous vehicles. However, the scarcity of labeled data limits performance on complex, newer datasets. To overcome this, semi-supervised learning allows a model to utilize unlabeled or under-labeled image sets through pseudo-labeling, improving accuracy and robustness without requiring dense annotation.

Getting around the problem of labeling is both interesting and extremely relevant to the current fast-growing AI industry. Classifying masks without pre-labeled images on an entire dataset has strong potential to generalize to other domains, such as autonomous vehicles, where exhaustive annotation is infeasible.

Convolutional neural networks, particularly the U-Net architecture [1], remain dominant for segmentation tasks due to their ability to capture fine spatial detail through encoder-decoder structures with skip connections. Vision Transformers (ViTs) [2] offer a complementary approach by modeling long-range dependencies through self-attention, providing stronger global context. This project combines both in a cross-teaching framework, where the two models exchange high-confidence pseudo-labels during training, aiming to leverage their complementary inductive biases under limited annotation.

## 2. RELATED WORK

Convolutional architectures such as U-Net [1] and its larger variants have demonstrated strong performance in biomedical segmentation. Transformer-based architectures introduced an alternative modeling approach through self-attention. The Vision Transformer (ViT) [2] laid the foundation for transformer-based vision models, while hybrid CNN-Transformer designs such as TransUNet demonstrated improved segmentation by combining local and global representations.

Semi-supervised segmentation has been explored through pseudo-labeling, consistency regularization, and self-ensembling. Cross-teaching and co-training frameworks extend these ideas by allowing multiple models to exchange pseudo-labels, reducing confirmation bias. Luo et al. [3] demonstrate that combining CNN and Transformer architectures in cross-teaching frameworks improves segmentation performance, motivating the combined U-Net and ViT approach evaluated here.

## 3. METHOD

### 3.1. Dataset and Problem Formulation

We use the Oxford-IIIT Pet segmentation dataset [4], downloaded from TensorFlow Datasets, which contains approximately 7,000 images of cats and dogs with pixel-level masks for pet, border, and background classes. To simulate a semi-supervised setting, only 20% of the labeled data was used as ground truth; the remaining images were treated as unlabeled during training. This setup enables direct comparison between supervised and semi-supervised performance.

### 3.2. Model Architecture

#### 3.2.1. U-Net

The first network is a U-Net with a ResNet-34 encoder pre-trained on ImageNet, implemented using the `segmentation_models_pytorch` library. The encoder is frozen during training so that optimization affects only the decoder and segmentation head. Its convolutional architecture is well suited to capture local structural patterns.

### 3.2.2. Vision Transformer (ViT) Segmentation Model

The second network is based on ViT-Base (patch16-224) from the `timm` library, operating on 3-channel  $224 \times 224$  inputs. The original classification head is replaced with a custom decoder of transposed convolution layers to reconstruct a dense segmentation map. The standard image resolution of 224 pixels with 16-pixel patches follows conventions from [2].

### 3.3. Semi-Supervised Cross-Teaching Framework

A mutual pseudo-labeling strategy is employed, inspired by the cross-teaching approach of Luo et al. [3]: each model acts as a teacher for the other by generating pseudo-labels on unlabeled samples.

#### 3.3.1. Supervised Learning on Labeled Data

For labeled batches, both networks are trained independently with ground-truth masks using cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c})$$

Parameters are updated via Adam (learning rate  $1 \times 10^{-4}$ ).

#### 3.3.2. Unlabeled Learning via Cross-Teaching

For unlabeled samples, each model generates predictions that are used as pseudo-labels for the other. Pixel-wise confidence is computed as the maximum softmax probability; predictions exceeding a threshold of 0.9 are considered reliable. The ViT predictions are upsampled to  $512 \times 512$  for the U-Net, while U-Net predictions are downsampled to  $224 \times 224$  for the ViT. Each model is then trained on the pseudo-labels via a consistency loss weighted by 0.5:

$$\mathcal{L}_{\text{cons}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C \hat{y}_{i,c}^{(t)} \log p_{i,c}^{(s)}$$

This mechanism encourages mutually consistent segmentations and allows the unlabeled set to contribute to representation learning.

### 3.4. Ensemble Inference

During inference, predictions from both models are combined by averaging their probability maps:

$$\hat{y}_{\text{ensemble}} = \frac{1}{2} \left( \text{softmax}(f_{\text{UNet}}(x)) + \text{softmax}(f_{\text{ViT}}(x)) \right)$$

The final mask is obtained by argmax over the ensemble output.

## 4. EXPERIMENTAL DESIGN

Both models were initialized from pre-trained encoders: the U-Net from `segmentation_models_pytorch` and the ViT from `timm`. Supervised transfer learning was first used to fine-tune the segmentation heads on the labeled portion of the Oxford Pet dataset [4], followed by joint semi-supervised training.

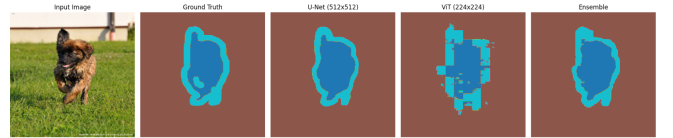
Both models use the Adam optimizer with a learning rate of  $1 \times 10^{-4}$ , a batch size of 8, and 30 training epochs. The confidence threshold for pseudo-label generation is 0.9 (averaged over the image), and the consistency loss is weighted by 0.5.

## 5. RESULTS

Table 1 shows the segmentation performance of all evaluated models. The U-Net [1] achieves a strong baseline, benefiting from its convolutional inductive bias on local structure. The ViT [2] scores lower on its own but contributes complementary global representations. Both cross-teaching models outperform their supervised stand-alone counterparts, and the ensemble achieves the highest scores across every metric, confirming that the two architectures provide useful, non-redundant signals.

|   | Model    | Dice     | IoU      | Pixel Acc |
|---|----------|----------|----------|-----------|
| 0 | U-Net    | 0.852017 | 0.763294 | 0.914292  |
| 1 | ViT      | 0.761595 | 0.649737 | 0.862519  |
| 2 | Ensemble | 0.852094 | 0.764581 | 0.919677  |

**Table 1.** Segmentation performance across models (Dice, IoU, Pixel Acc).



**Fig. 1.** Qualitative segmentation results: U-Net, ViT, and ensemble.

Qualitatively (Figure 1), the U-Net produces the smoothest boundaries visually, while the ViT introduces some block-like artifacts despite reasonable quantitative scores. The ensemble output combines the strengths of both, producing clean, consistent masks.

## 6. DISCUSSION

The cross-teaching framework [3] improved both architectures, though the impact differs. With the supervised U-Net baseline already strong, the absolute gain from cross-teaching is modest — however, the real advantage emerges in settings with larger, mostly unlabeled datasets where a supervised U-Net cannot train effectively. The ViT, despite lower stand-alone performance, still provides useful information for ensemble inference, as evidenced by the ensemble outperforming either model alone.

These results are consistent with standard hyperparameter choices: learning rate, batch size, and the  $224 \times 224$  ViT resolution all follow established conventions [2]. The ViT decoder may benefit from additional context during upsampling, which could be addressed with a more expressive decoder in future work. Compute constraints also limited hyperparameter search; faster training would enable more extensive ablations.

## 7. CONCLUSION

This study shows that semi-supervised cross-teaching [3] is an effective strategy for improving segmentation when labeled data is limited. The ensemble cross-teaching model outperforms both individual models across all metrics, demonstrating that a U-Net [1] and ViT [2] provide complementary representations that benefit from mutual pseudo-label exchange. While the supervised U-Net baseline is competitive on this dataset, the cross-teaching framework is most valuable in larger, under-labeled settings where supervised training is infeasible.

Future work includes testing on more challenging and deeper datasets, exploring detection decoder heads (which was already started but not added due to timing and compute constraints), and investigating the impact of confidence threshold and consistency weight as hyperparameters.

## 8. REFERENCES

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